



University of Al-Ameed
College of Medicine
Department of Biochemistry



Protein Digestion

Assistant prof.
Dr. WEAAM AWAD

Objectives

To understand the following topics:

- Digestion of **proteins**
- Absorption of **amino acids**
- **Meister cycle**

Digestion in the Stomach

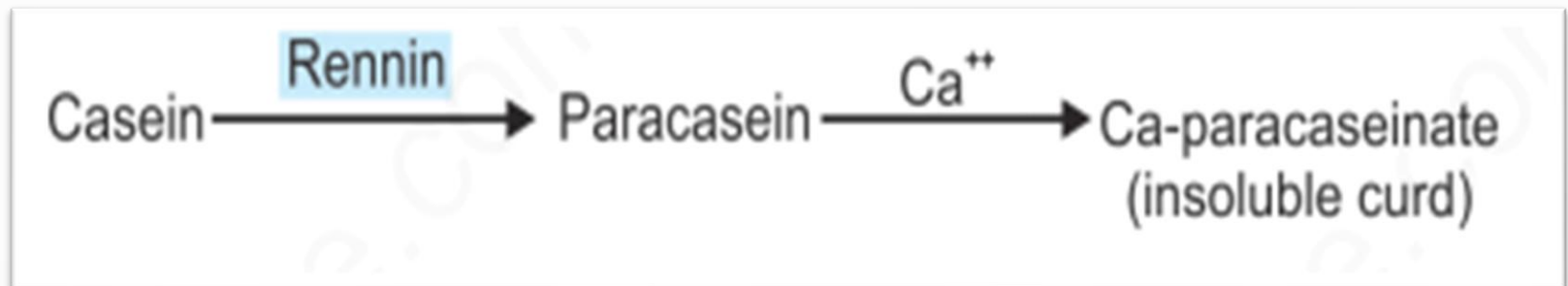
When protein enters the stomach, it stimulates the secretion of the hormone gastrin, from gastric mucosal cells, which in turn, stimulates the release of gastric juice containing **hydrochloric acid**, proenzyme **pepsinogen** and **rennin** in infants.

1- Hydrochloric acid

- Secreted by The gastric parietal cells
- It makes the pH optimum for the action of pepsin and also activates pepsin.
- It denatures the protein

2- Rennin

Action of rennin is to clot milk. This is accomplished by the slight hydrolysis of the casein of milk to produce **paracasein**, which coagulates in the presence of calcium ions, resulting in an insoluble **calcium-paracaseinate** curd. Calcium paracaseinate is then acted on by pepsin. The purpose of this reaction is to convert milk into a more solid form to prevent the rapid passage of milk from the stomach of infant



3- Pepsin

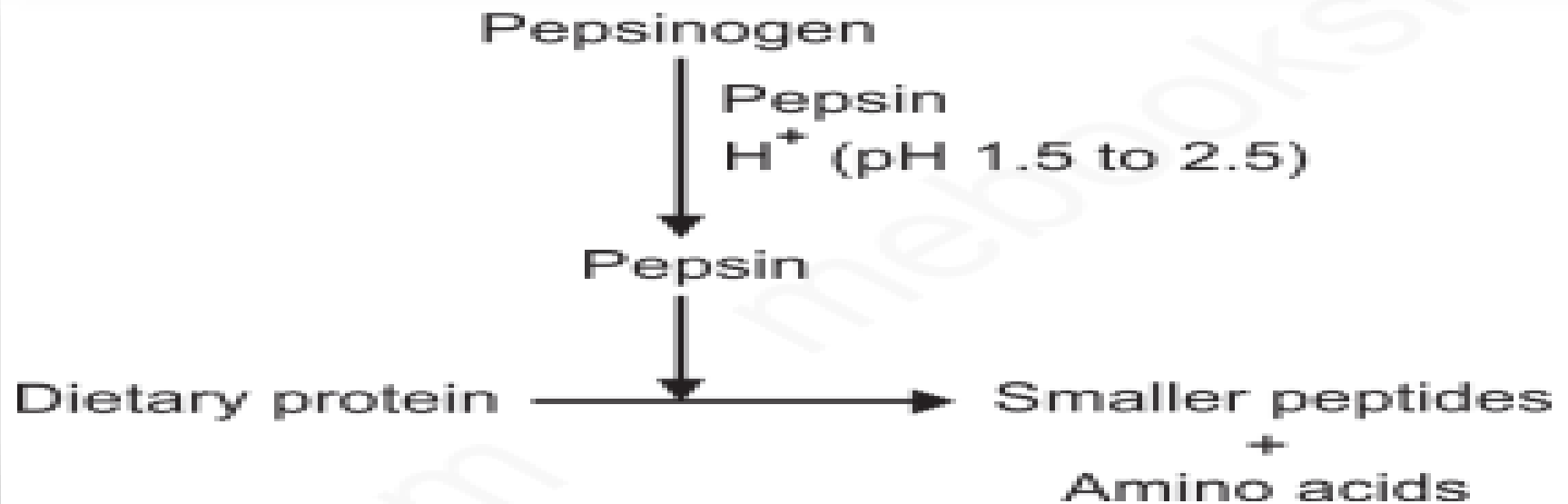
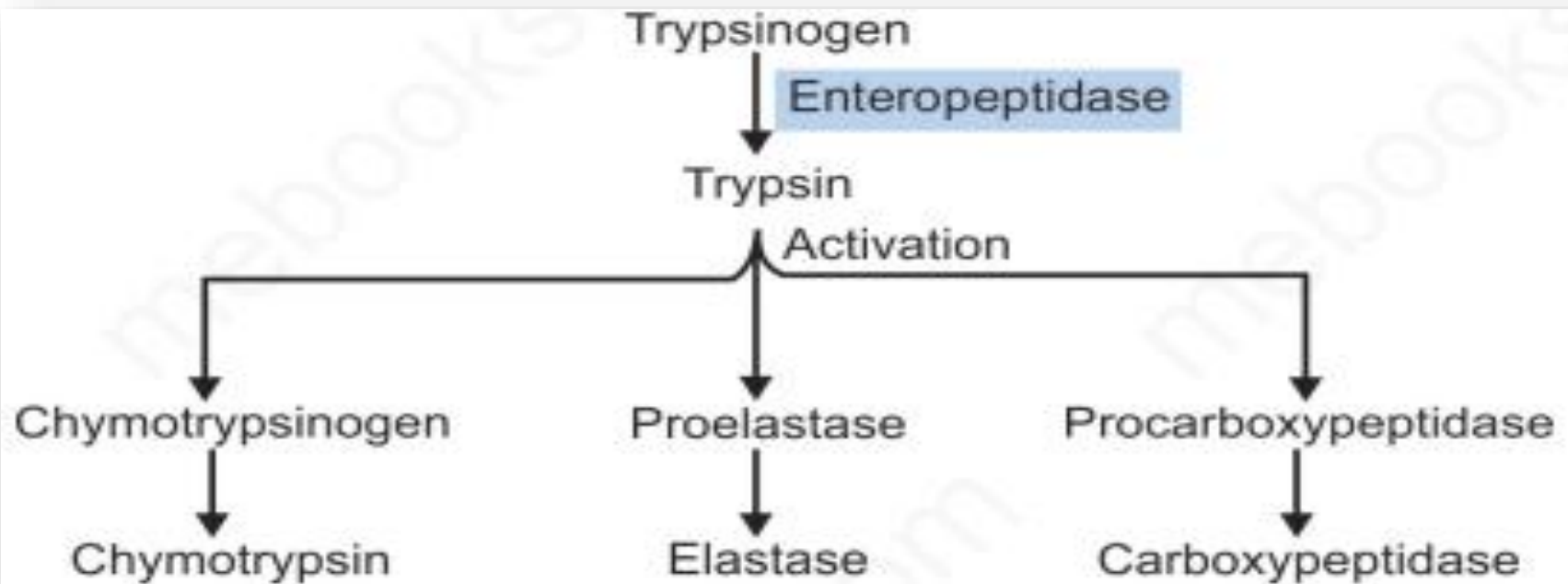
- ✓ It is secreted as the proenzyme pepsinogen, an inactive form
- ✓ It is converted into active pepsin in the gastric juice by the enzymatic action of pepsin itself or by high hydrogen ion concentration
- ✓ Pepsin catalyses hydrolysis of peptide bonds adjacent to amino acids with bulky side chains like : Aromatic amino acids and Branched amino acids .Thus, pepsin cleaves long polypeptide chains into a mixture of smaller peptides and some free amino acids

Pancreatic Digestion of Proteins

The optimum pH for the activity of pancreatic enzymes (pH 8) , Pancreatic juice contains the important endopeptidases

- Trypsin
- Chymotrypsin
- Elastase
- Carboxypeptidase.

These enzymes are also secreted as inactive zymogens which are converted to their active form in the intestinal lumen. This would prevent autodigestion of the pancreatic secretory acini



Intestinal Digestion of Proteins

Complete digestion of the small peptides to the level of amino acids is brought about by enzymes present in intestinal juice. The luminal surface of intestinal epithelial cell contains the following enzymes:

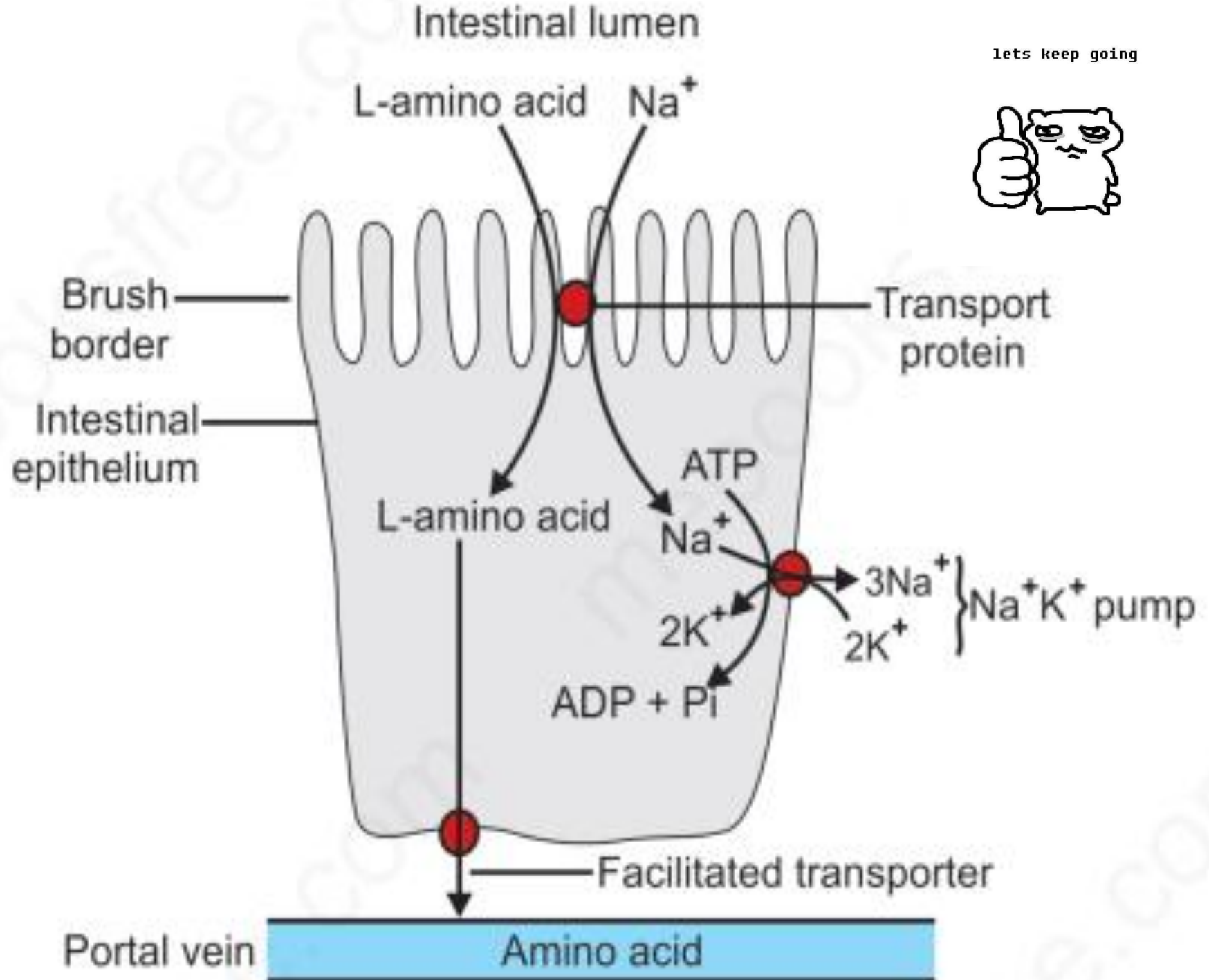
- **aminopeptidase**
- **Dipeptidases and tripeptidases**

They will bring about the complete digestion of proteins

ABSORPTION OF AMINO ACIDS

The absorption of most amino acids involves an active transport mechanism, requiring ATP and specific transport proteins in the intestinal mucosal cells, glutathione participates in absorption of amino acids in intestine, kidneys and brain and the cycle is called gamma-glutamyl cycle or Meister cycle

lets keep going



Q Determine the location within the human body where each of the following aspects of protein digestion occurs?

- a. The enzyme trypsin is active.
- b. Large polypeptides are produced.
- c. Protein is denatured by HCl.
- d. Active transport moves amino acids into the bloodstream



SUMMARY

Proteolytic enzymes (also called proteases) break down dietary proteins into their constituent amino acids in the stomach and the intestine. Many of these digestive proteases are synthesized as larger, inactive forms known as zymogens.

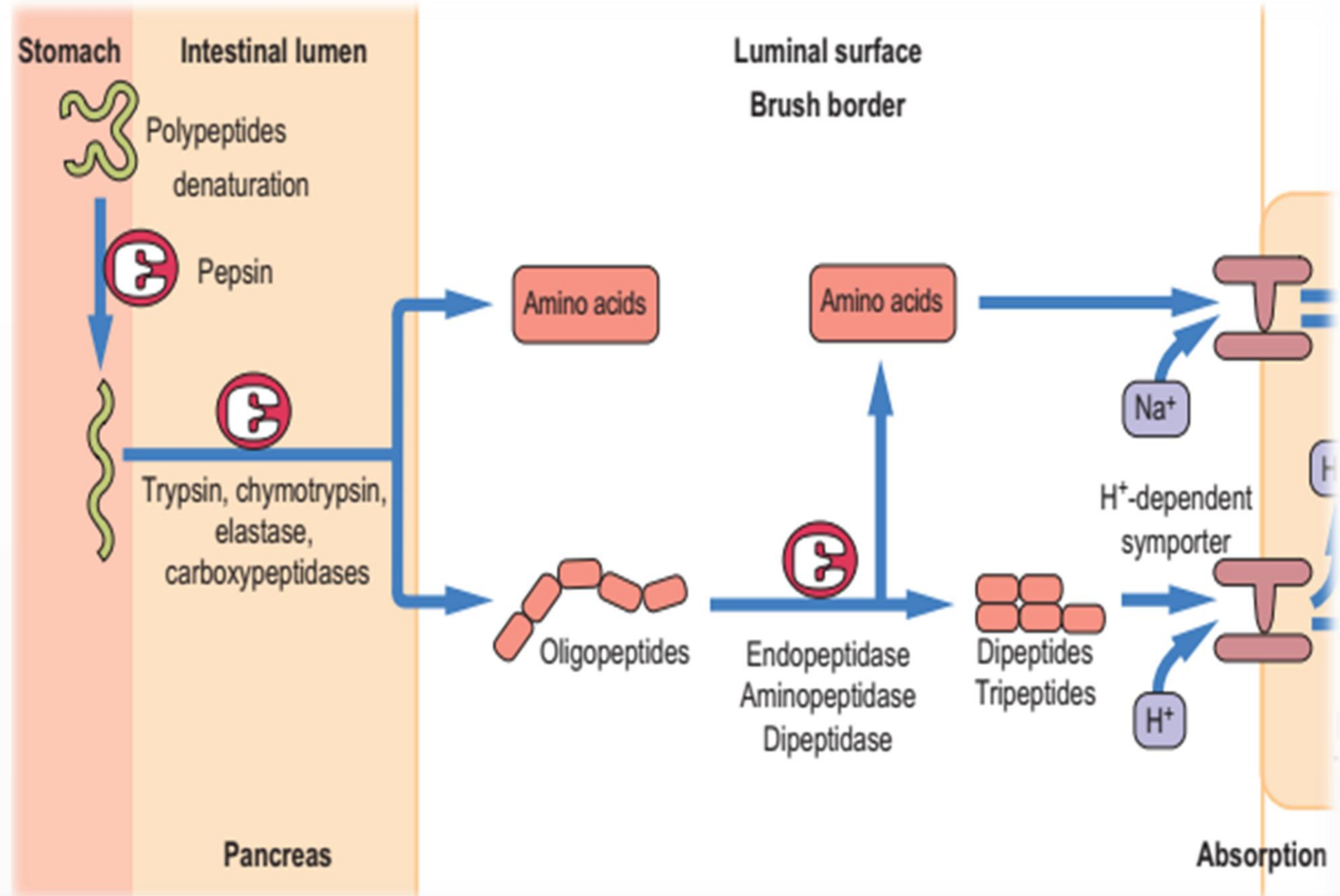
After zymogens are secreted into the digestive tract, they are cleaved to produce the active proteases.

In the stomach, **pepsin** begins the digestion of proteins by hydrolyzing them to smaller polypeptides.

The contents of the stomach pass into the small intestine, where enzymes produced by the exocrine pancreas act. The pancreatic proteases (**trypsin**, **chymotrypsin**, **elastase**, and the **carboxypeptidases**) cleave the polypeptides into oligopeptides and amino acids

Further cleavage of the oligopeptides to amino acids is accomplished by enzymes produced by the intestinal epithelial cells. These enzymes include **aminopeptidases** located on the brush border and other peptidases located within the cells. Ultimately, the amino acids produced by protein digestion are absorbed through the intestinal epithelial cells and enter the blood

<i>Locations</i>	<i>Agents</i>	<i>Outcomes</i>
Stomach Stomach	Acid (HCl) Pepsin	Denaturation Large peptides + some free amino acids
Small intestine lumen (pancreatic enzymes)	Trypsin chymotrypsin elastase carboxy- peptidases	Oligopeptides (2-10 amino acids) + free amino acids
Brush border surface of intestine	Endopepti- dases and aminopepti- dases	Di- or tri-peptides + free amino acids



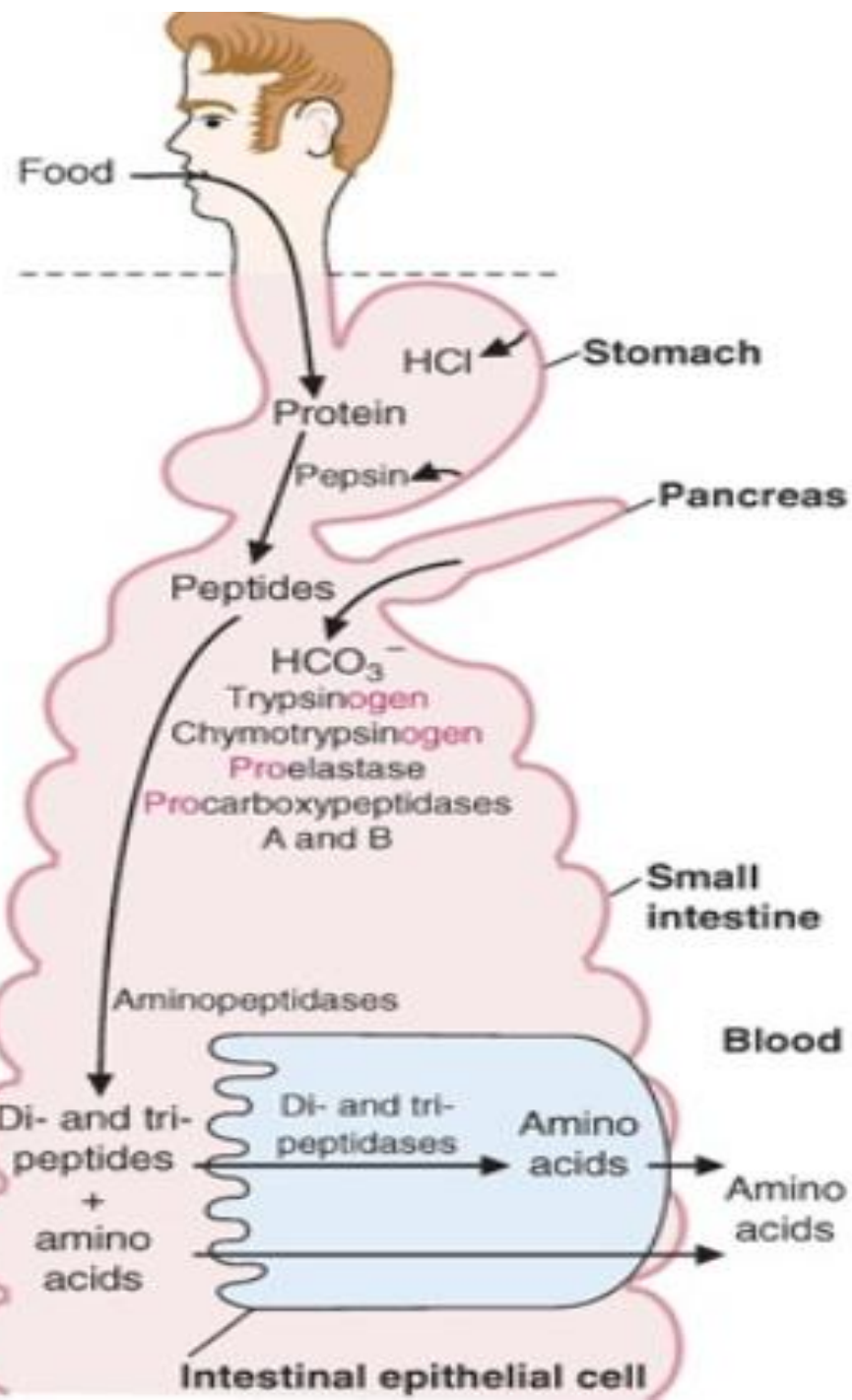
1. The zymogen form trypsinogen of pancreatic juice is converted to 'active' trypsin by:

- a. Pepsin
- b. Enterocrinin
- c. Enterokinase
- d. Rennin
- e. Gastrin

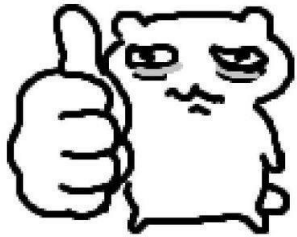
2. Inactive zymogens (proenzymes) are precursors of all the following gastrointestinal enzymes, except:

- a. Trypsin
- b. Carboxypeptidase
- c. Pepsin
- d. Aminopeptidase
- e. Chymotrypsin

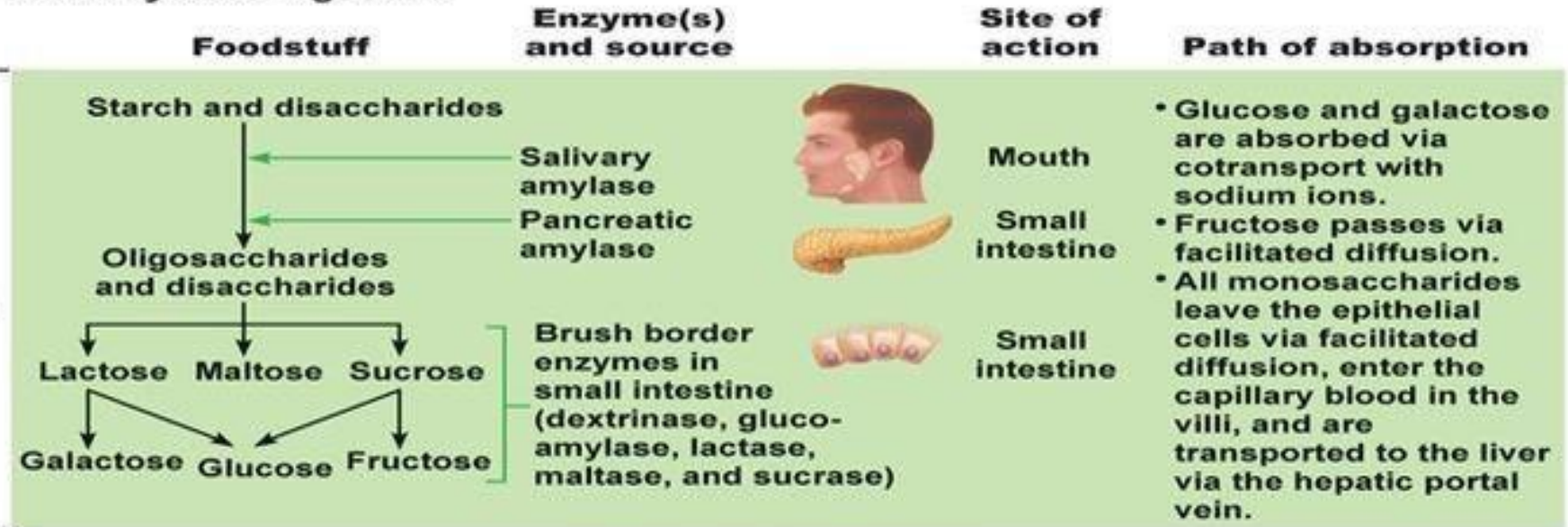




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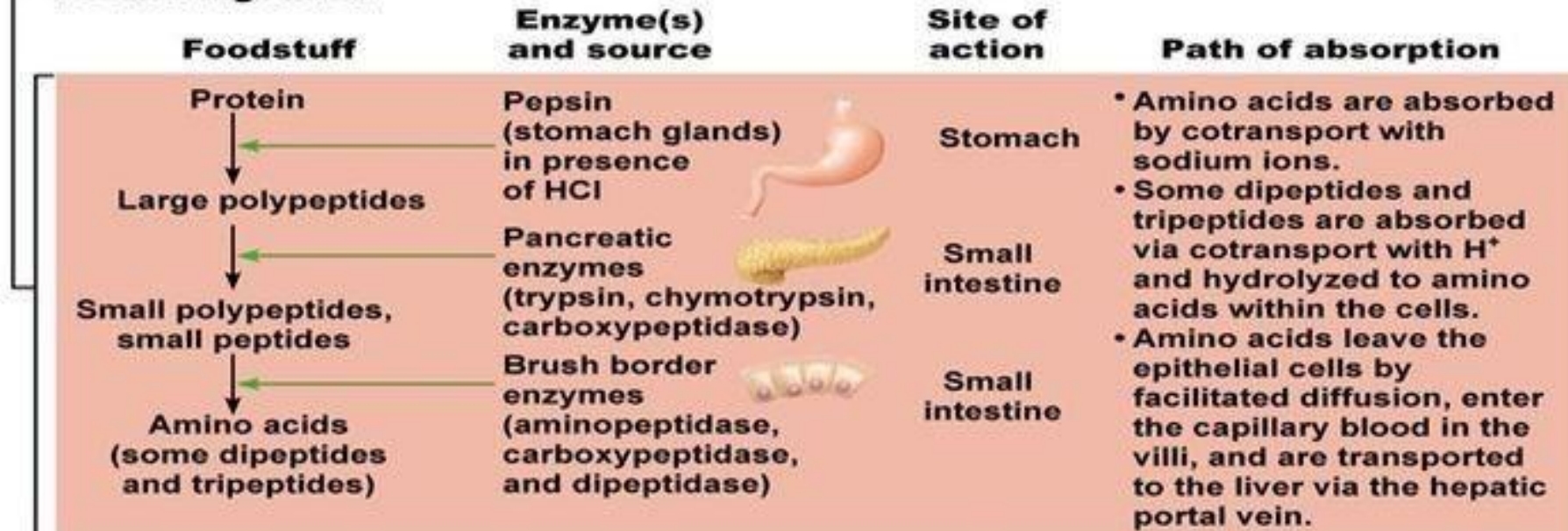


Carbohydrate digestion



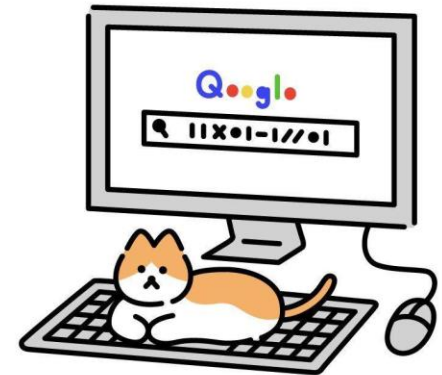
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Protein digestion



Homework

Disorders of protein absorption



THANK YOU

